Spectral Theory of Isogeny Graphs and Quantum Sampling of Secure Supersingular Elliptic Curves

Maher Mamah

joint work with Jake Doliskani and David Jao

Department of Combinatorics and Optimization
University of Waterloo

UNIVERSITY OF WATERLOO

Isogeny Club
Season 9

NSERC CRSNG

“ Our quest to generate authentic banknotes led us to the problem of sampling supersingular elliptic curves with unknown endomorphism rings.."

SECTION 1

Why bother?



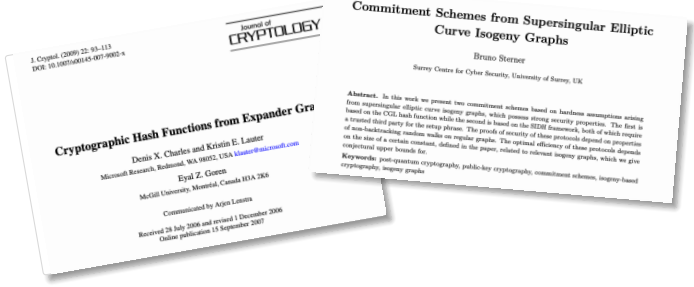
J. Cryptol. (2009) 22: 93–113
DOI: 10.1007/s00145-007-9002-x

Journal of CRYPTOLOGY

Cryptographic Hash Functions from Expander Graphs

Denis X. Charles and Kristin E. Lauter
Microsoft Research, Redmond, WA 98052, USA klauter@microsoft.com
Eyal Z. Goren
McGill University, Montréal, Canada H3A 2K6

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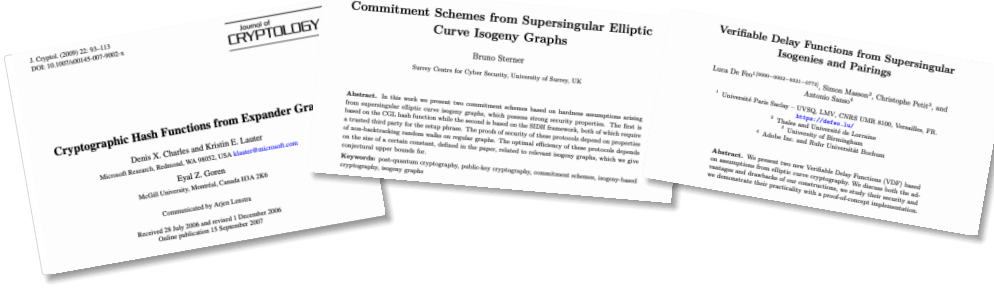
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Commitment Schemes from Supersingular Elliptic Curve Isogeny Graphs

Bruno Sterner
Survey Centre for Cyber Security, University of Surrey, UK

Abstract. In this work we present two commitment schemes based on hardness assumptions arising from supersingular elliptic curve isogeny graphs, which possess strong security properties. The first is based on the CGL hash function while the second is based on the HDM framework, both of which require a trusted third party for the setup phrase. The proofs of security of these protocols depend on properties of non-backtracking random walks on regular graphs. The optimal efficiency of these protocols depends on the size of a certain constant, defined in the paper, related to ramanujan isogeny graphs, which we give conjectural upper bounds for.

Keywords: post-quantum cryptography, public-key cryptography, commitment schemes, isogeny-based cryptography, isogeny graphs



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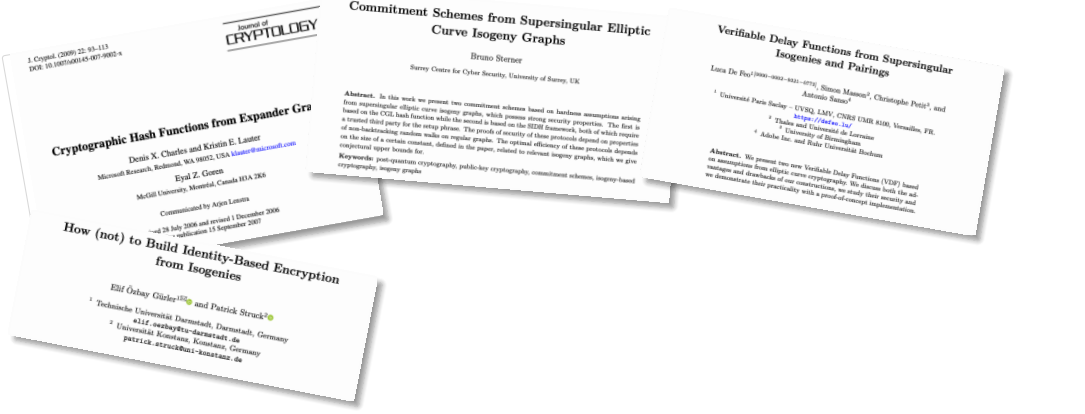
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Verifiable Delay Functions from Supersingular Isogenies and Pairings

Luca De Feo[1][0000−0002−9321−0779], Simon Masson[2], Christophe Petit[3], and Antonin Sanso[4]
[1] Université Paris Saclay – UVSQ, LMV, CNRS UMR 8100, Versailles, FR.
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How (not) to Build Identity-Based Encryption from Isogenies

Elif Özbay Gürler[1][0] and Patrick Struck[2][0]
[1] Technische Universität Darmstadt, Darmstadt, Germany
elif.oezbay@tu-darmstadt.de
[2] Universität Konstanz, Konstanz, Germany
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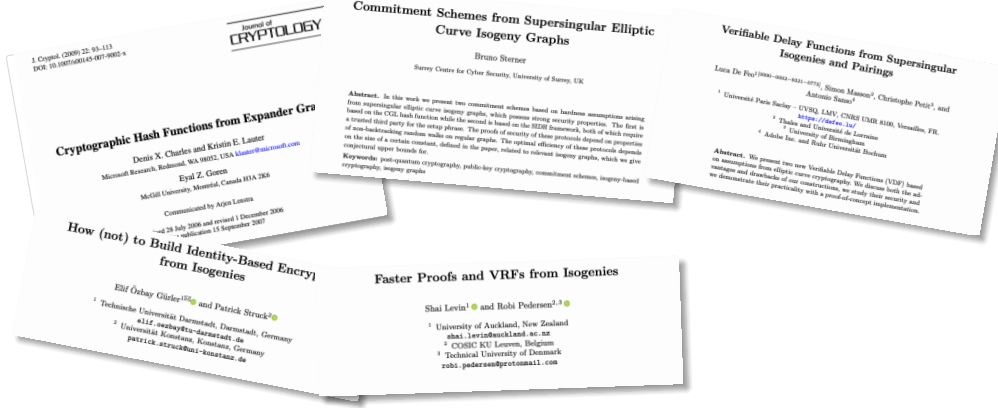
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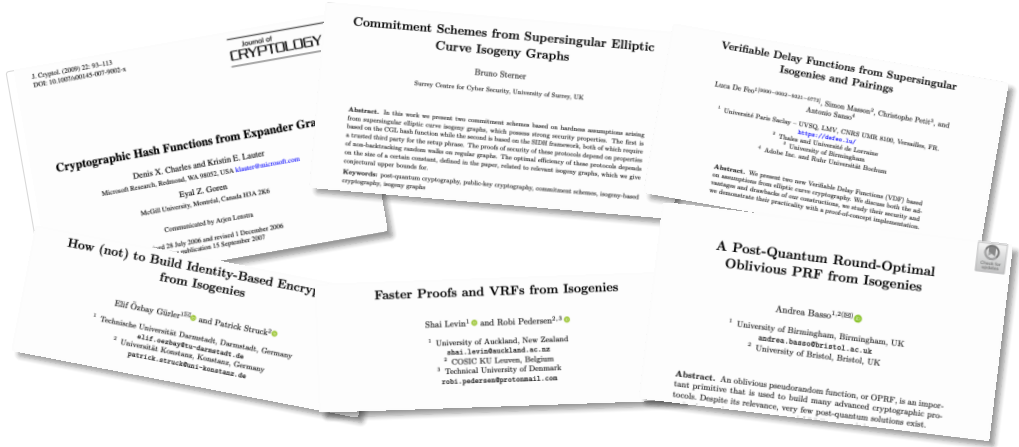
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Faster Proofs and VRFs from Isogenies

Shai Levin[1] and Robi Pedersen[2,3]
[1] University of Auckland, New Zealand
shai.levin@auckland.ac.nz
[2] COSIC KU Leuven, Belgium
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robi.pedersen@protonmail.com



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shai.levin@auckland.ac.nz
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robi.pedersen@protonmail.com

A Post-Quantum Round-Optimal Oblivious PRF from Isogenies

Andrea Basso[1,2][0000]
[1] University of Birmingham, Birmingham, UK
andrea.basso@bristol.ac.uk
[2] University of Bristol, Bristol, UK

Abstract. An oblivious pseudorandom function, or OPRF, is an important primitive that is used to build many advanced cryptographic protocols. Despite its relevance, very few post-quantum solutions exist.



No exposed endomorphism ring

$\mathrm{End}(E_0)$ must remain hidden from EVERYONE.

SECTION 2

Previous attempts

Bröker's Algorithm

Goal: Given a prime p, construct a supersingular elliptic curve $E/\mathbb{F}_{p^2}$.

1. Find a small prime $q \equiv 3 \pmod 4$ such that

$$\left(\frac{-q}{p}\right) \neq 1.$$

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$$\Delta_{\mathcal{O}} = -q \qquad \text{(or } -4q\text{, depending on } q\text{)}.$$

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$$j \in \overline{\mathbb{F}}_p \qquad \text{such that} \qquad H_{\mathcal{O}}(j) = 0.$$
5. Construct an elliptic curve E with $j(E) = j$. By Deuring reduction, E is supersingular.

Limitations

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- However, $\mathrm{End}(E)$ will contain the suborder $\mathcal{O}$ with unusually small degree endormorphism of order $O(q) = \mathrm{polylog}(p)$, we can compute the whole of $\mathrm{End}(E)$ [LB20].

Limitations

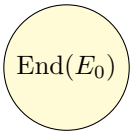
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What if we take a random walk in the isogeny graph?

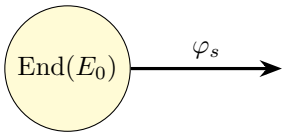
Known endomorphism ring

ℓ-IsogenyPath $\longleftrightarrow$ EndRing



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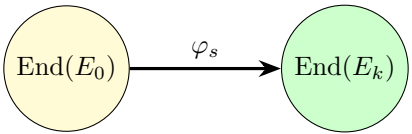


Compute the isogeny path

$$\varphi_s : E_0 \to E_k$$

Known endomorphism ring

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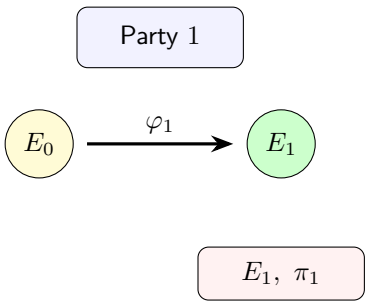
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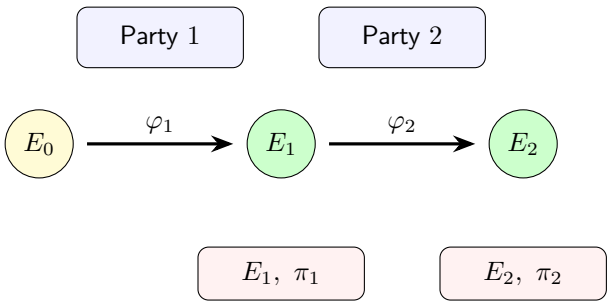
Trusted Setup of Basso et al.



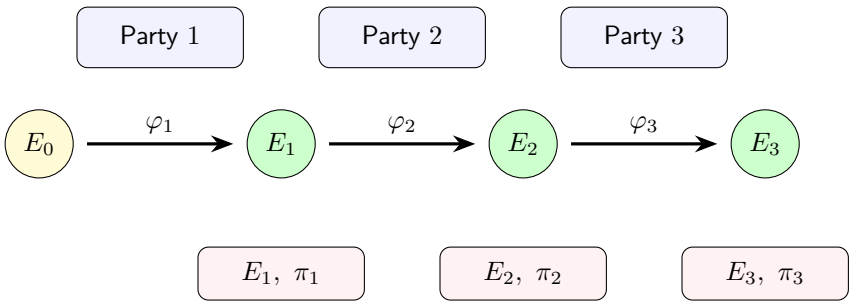
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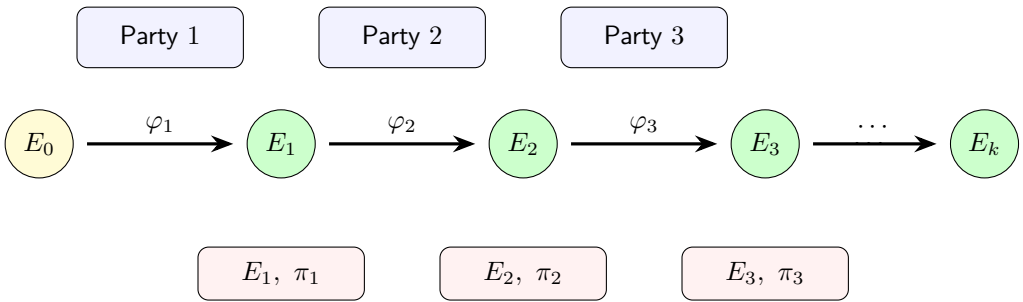
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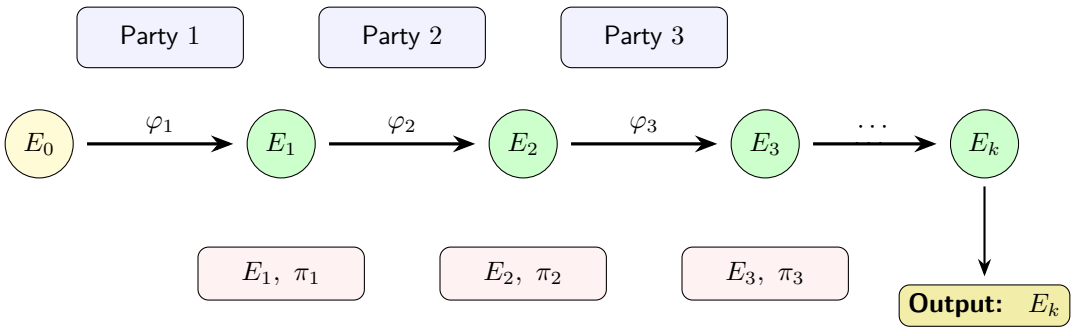
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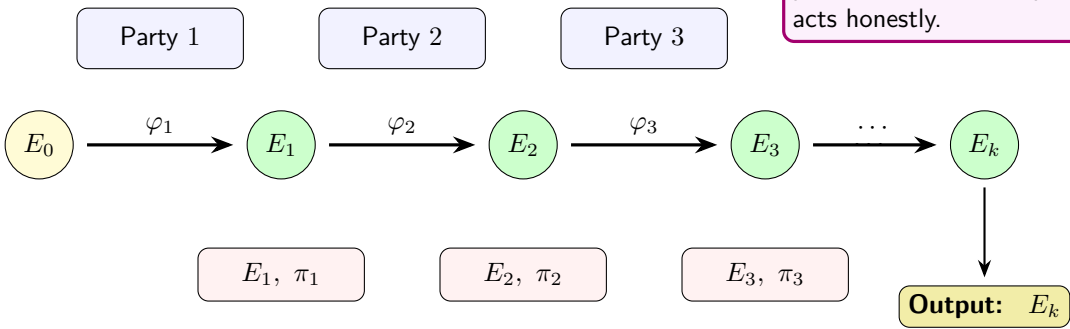


Trusted Setup of Basso et al.



Trusted Setup of Basso et al.

Key point:
The output curve is secure provided at least one party acts honestly.



OXFORD

The Computer Journal, 2024, **67**, 2702–2719
https://doi.org/10.1093/comjnl/bxae038
Advance access publication date 24 May 2024
Original Article

Failing to Hash Into Supersingular Isogeny Graphs

Jeremy Booher[1], Ross Bowden[2], Javad Doliskani[3], Tako Boris Fouotsa[4], Steven D. Galbraith[5], Sabrina Kunzweiler[6], Simon-Philipp Merz[7], Christophe Petit[8,9], Benjamin Smith[10], Katherine E. Stange[11,*], Yan Bo Ti[12], Christelle Vincent[13], José Felipe Voloch[14], Charlotte Weitkämper[15] and Lukas Zobernig[5]

[1]Department of Mathematics, University of Florida, Gainsville, FL 32611, USA
[2]School of Computer Science, University of Bristol, Bristol, BS8 1UB, UK
[3]Department of Computing and Software, McMaster University, Hamilton, L8S 4L7, Canada
[4]LASEC, EPFL, 1015 Lausanne, Switzerland
[5]Department of Mathematics, The University of Auckland, Auckland, 1010, New Zealand
[6]Ruhr-Universität Bochum, 44801 Bochum, Germany
[7]Department of Computer Science, ETH Zurich, 8092 Zürich, Switzerland
[8]Laboratoire d'Informatique, Université libre de Bruxelles, 1050 Bruxelles, Belgium
[9]University of Birmingham, Birmingham, B15 2TT, UK
[10]Inria and Laboratoire d'Informatique (LIX), CNRS, École polytechnique, Institut Polytechnique de Paris, 91120 Palaiseau, France
[11]Department of Mathematics, University of Colorado Boulder, Boulder, CO 80309, USA
[12]DSO National Laboratories, 118225, Singapore
[13]Department of Mathematics and Statistics, University of Vermont, Burlington, VT 05405, USA
[14]School of Mathematics and Statistics, University of Canterbury, Christchurch, 8140, New Zealand
*Corresponding author: ketange@math.colorado.edu

An important open problem in supersingular isogeny-based cryptography is to produce, without a trusted authority, concrete examples of 'hard supersingular curves' that is equations for supersingular curves for which computing the endomorphism ring is as difficult as it is for random supersingular curves. A related open problem is to produce a hash function to the vertices of the supersingular ℓ-isogeny graph, which does not reveal the endomorphism ring, or a path to a curve of known endomorphism ring. Such a hash function would open up interesting cryptographic applications. In this paper, we document a number of (thus far) failed attempts to solve this problem, in the hope that we may spur further research, and shed light on the challenges and obstacles to this endeavour. The mathematical approaches contained in this article include: (i) iterative root-finding for the supersingular polynomial; (ii) ged's of specialized modular polynomials; (iii) using division polynomials to create small systems of equations; (iv) taking random walks in the isogeny graph of abelian surfaces, and applying Kummer surfaces and (v) using quantum random walks.

Keywords: isogeny-based cryptography; hashing; elliptic curves.

On random sampling of supersingular elliptic curves

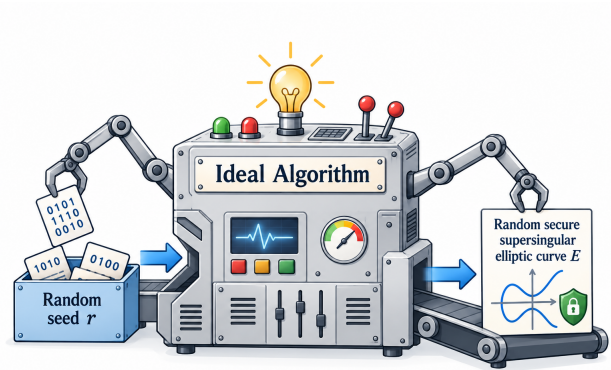
Marzio Mula[1] · Nadir Murru[2] · Federico Pintore[2]

Received: 8 March 2024 / Accepted: 13 November 2024 / Published online: 2 December 2024
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Abstract

We consider the problem of sampling random supersingular elliptic curves over finite fields of cryptographic size (SRS problem). The currently best-known method combines the reduction of a suitable complex multiplication (CM) elliptic curve and a random walk over some supersingular isogeny graph. Unfortunately, this method is not suitable when the endomorphism ring of the generated curve needs to be hidden, like in some cryptographic applications. This motivates a stricter version of the SRS problem, requiring that the sampling algorithm gives no information about the endomorphism ring of the output curve (cSRS problem). In this work we formally define the SRS and cSRS problems, which are both of theoretical interest. We discuss the relevance of the two problems for cryptographic applications, and we provide a self-contained survey of the known approaches to solve them. Those for the cSRS problem have exponential complexity in the characteristic of the base finite field (since they require computing and finding roots of polynomials of large degree), leaving the problem open. In the second part of the paper, we propose and analyse some alternative techniques—based either on the Hasse invariant or division polynomials—and we explain the reasons why they do not readily lead to efficient cSRS algorithms, but they may open promising research directions.

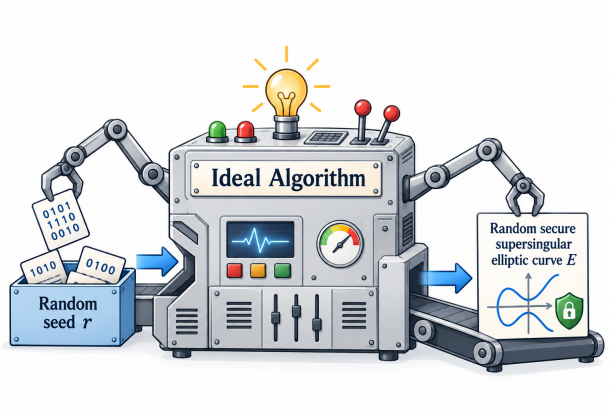
Ideal algorithm



Ideal algorithm

The remaining issue

I still need to prove that I executed the algorithm correctly.

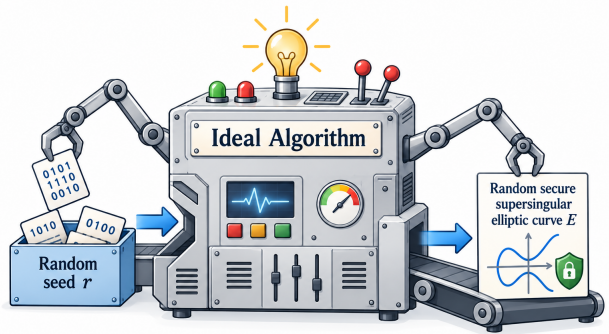


Ideal algorithm

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I still need to prove that I executed the algorithm correctly.

There are classical verifiable computation protocols which do the job, Pinnochio e.g. [PHGR13].



SECTION 3

Spectral theory of Isogeny graphs

Two views of the same object

Geometric world

$\mathcal{G}_\ell(p)$: ℓ-Isogeny Graph

- Vertices: supersingular elliptic curves

$$E/\mathbb{F}_{p^2}.$$

- Edges: degree-ℓ isogenies

$$E \longrightarrow E'.$$

- Walks correspond to chains of isogenies.

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Deuring correspondence: Isogeny Graphs $\longleftrightarrow$ Brandt Graphs

Adjacency matrices and eigenvectors

The adjacency matrix

Let

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Thus the eigenvector assigns an amplitude

$$\alpha_i = \phi(E_i)$$

to every vertex E_i.

The Hecke eigenbasis

A family of adjacency operators

For distinct primes

$$\ell_1, \ldots, \ell_r \neq p,$$

consider the corresponding adjacency matrices

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Hence there exists a joint orthonormal eigenbasis

$$\boxed{\{\phi_1, \ldots, \phi_n\} \quad \forall A_{\ell_i}}$$

The sup-norm problem and delocalization

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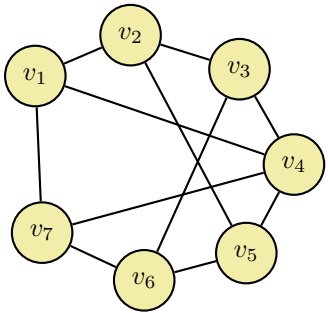
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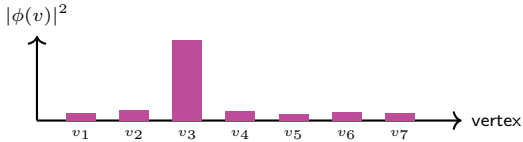
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Localization vs. delocalization

Underlying graph

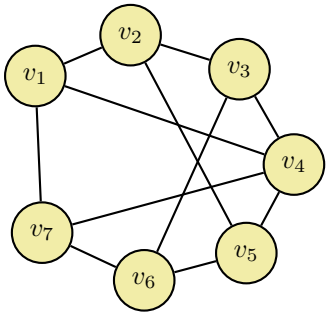


Localized

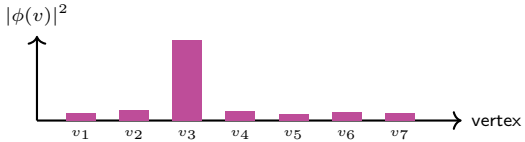


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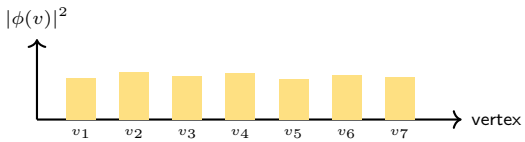
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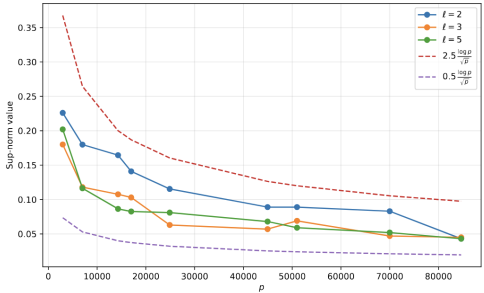


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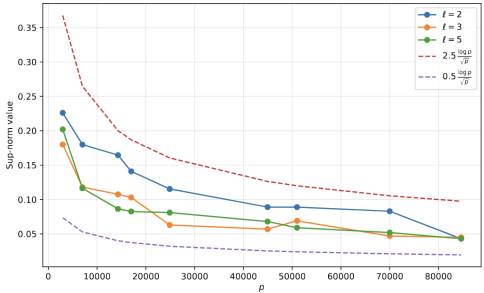


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Spoiler: This will be crucial for proving the hardness of the output curves.

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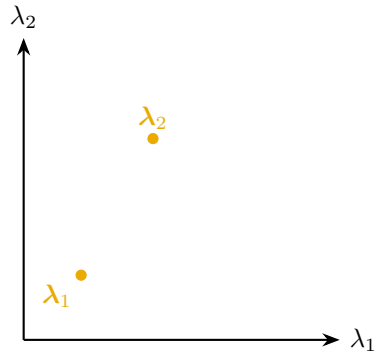
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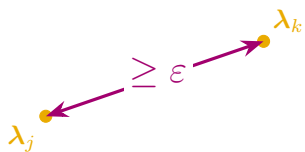
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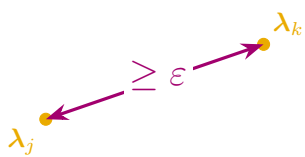


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Why separation matters

Under ε-separation, each spectral tag $\boldsymbol{\lambda}_j$ **uniquely identifies** its eigenvector ϕ_j.

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Our result under GRH

- Consider all primes $2 \leq \ell_i = O(\log^4 p)$.
- For any two distinct **non-trivial simultaneous eigenvectors**, we prove

$$\boxed{\varepsilon \gg 1}.$$

Heuristic separation

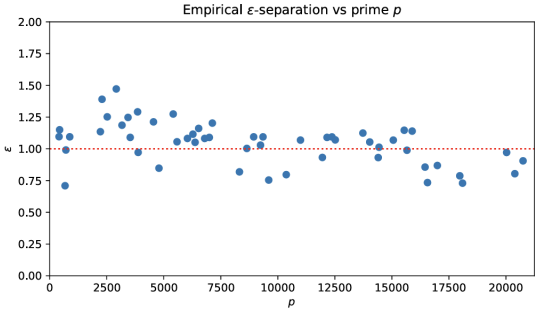


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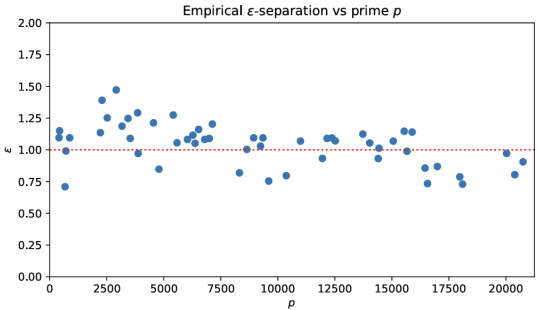


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SECTION 4

Quantum Sampling from the Full Isogeny Graph

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For instance, any supersingular elliptic curve vertex can be encoded by the Weirstrass coefficients (a, b) such that $y^2 = x^3 + ax + b$.

$$|(a, b)\rangle, \quad a, b \in \mathbb{F}_{p^2}$$

Quantum Phase Estimation

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- Measuring the **eigenvalue register** filters the state according to its eigenvalue.

$$\boxed{\text{QPE} + \text{measurement} \implies \text{spectral filtering}}$$

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- Repeat the same procedure for

$$\ell_2, \ldots, \ell_r.$$

At step k, apply QPE with respect to $H_{\ell_k} = e^{i\frac{A_{\ell_k}}{\sqrt{\ell_k}}}$ and measure the eigenvalue register.

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Complexity of Algorithm 1

	primes used	quantum gate complexity
Heuristic	all $\ell_i \in [\log p,\, 4 \log p]$	$O(\log^{6.5} p)$
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P.S. **Many thanks** to Alessandro and the TUM research group for pointing out a flaw in an earlier version of the analysis, which we have since corrected.

Unconditional bound on Distribution

Theorem 7 (M.–Jao–Doliskani)

For every $\varepsilon > 0$ and every supersingular elliptic curve $E/\mathbb{F}_{p^2}$,

$$\Pr[\text{Algorithm 1 outputs } E] \ll_\varepsilon p^{-1+\varepsilon}.$$

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What about the sequence of measured eigenvalues $(\tilde{\lambda}_1, \ldots, \tilde{\lambda}_r)$?

Are the sampled curves secure?

Main Result

Assuming the sup-norm conjecture $\|\phi\|_\infty^2 \ll \log^2 p/p$,

The EndRing of a sampled curve is essentially as hard as for a uniformly random supersingular curve.

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Quantitatively – Theorem 8 of M., Jao and Doliskani

Any QPT attack against the output of $\mathrm{Algorithm}\ 1$ can be converted into an attack against $\textsc{EndRing}$ on a uniformly random curve, for $\alpha = O(\log^2 p)$

$$\Pr[\text{attack succeeds}] \ \leq \ \alpha \cdot \Pr[\text{uniform EndRing attack succeeds}] + O\left(\frac{\alpha}{\sqrt{p}}\right),$$

Why are the sampled curves secure?

Spectral tag

QPE reveals

$$\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_r).$$

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$$\Pr[E = y \mid \Lambda = \boldsymbol{\lambda}] \ll \frac{\log^2 p}{p}$$

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Security transfer

Uniform sampling gives

$$U(y) = \frac{1}{m}.$$

Our sampler satisfies

$$\boxed{P_{\boldsymbol{\lambda}}(y) \leq \alpha\, U(y)}.$$

Hence, the desired bound, with $\alpha = O(\log^2 p)$.

SECTION 5

Sampling Oriented curves

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Hence, fixing any $x_0 \in \mathcal{X}$,
$$g \longmapsto g * x_0$$
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- The action has a natural **Fourier eigenbasis**: for every $h \in G$,
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- For $k \in G$, define the **action operator**
$$U_k\,|y\rangle = |k * y\rangle\,.$$
The Fourier states are simultaneous eigenstates:
$$\boxed{U_k\,|G^{(h)} * x_0\rangle = \chi(-k,h)\,|G^{(h)} * x_0\rangle\,.}$$

Ideal world

$$\frac{1}{\sqrt{|G|}} \sum_{g \in G} |g \star x\rangle$$

The Quantum Sampling trick

- Using phase kickback, Compute.Index recovers the Fourier label:

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- We pick $\kappa(\cdot)$ to be a Quadratic refinement. Hence measuring gives a

$$\boxed{\text{uniformly random } g * x \in \mathcal{X}.}$$

Uniform sampling of oriented curves

Main theorem

Let $(\mathrm{Cl}(\mathcal{O}), \mathcal{X}, *)$ be a regular action on the set of $\mathcal{O}$-oriented supersingular elliptic curves over $\mathbb{F}_{p^2}$. Then there exists an algorithm that outputs a **uniformly random** and **secure**

$$(E, \iota) \in \mathcal{X}$$

in **quantum polynomial time** in

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- **Future work?**

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Thank you for your attention!


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